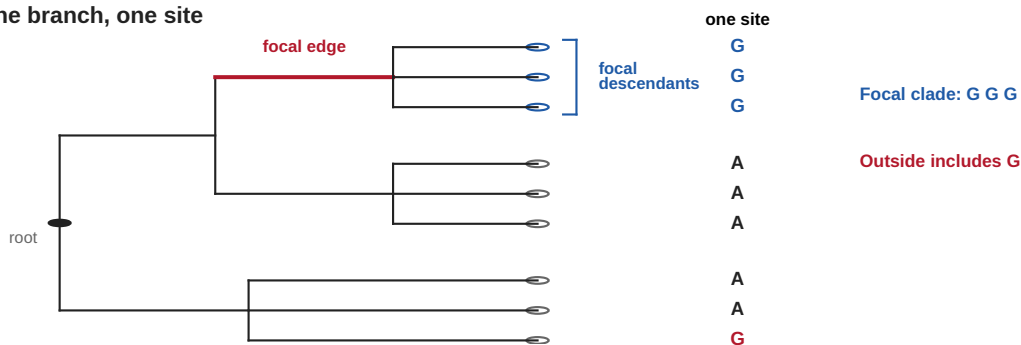


A One branch, one site



B Two questions, different answers

OBSERVED

Clade-exclusive?

focal G G G

outside A A A A A G

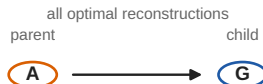
NO not fixed-exclusive
G occurs outside the clade

≠

kept separate

INFERRED

Substitution on the focal edge?



YES unambiguous A → G
Only optimal focal-edge pair: A → G

C Retain all optimal focal-edge solutions

Optimal parent → child pair(s)

BRANCHSNV reports

A → G only	Unambiguous change
A → G or C → G	State ambiguity
A → G or G → G	Placement ambiguity
G → G only	No change

No arbitrary tie-breaking

All globally optimal focal-edge state pairs are retained.

DESIGN SAFEGUARDS

Explicit root
defines direction

Exact descendants
defines branch

Exact taxon labels
defines correspondence

Complete optimal set
retains uncertainty

Deterministic provenance
supports auditability